Supplementary index

The exact maximum growth factor for complete pivoting on real 5×5 matrices

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Proof objects and reproducibility index

This index identifies the mathematical modules used in the article, their archived proof objects, and the scopes of the recorded exact acceptances. It is a map to the archived arguments, code, and large certificate payloads, not a replacement for them. Historical labels beginning R, V, B, or Round identify archived sources; they are not additional mathematical hypotheses. File and collection names identify manifest objects. The public repository is https://github.com/hkjtsgmc79-boop/rho5-proof; fixed release `v1.0.1` supplies the downloadable archives and verification guide. Relative locations refer to the companion analytical collection or to the named extracted archive, as specified in S8. Hashes identify bytes; mathematical validity depends on the analytic arguments, exact acceptances, and source-composition rules. Large certificate payloads, including transport parts for the historical-root archive, are supplied as release attachments. The repository documents exact sizes, checksums and computing requirements.

Reading this revision

The journal submission comprises the main article, the separate *Full proof supplement* (FP), and this index. The main article gives the complete lift, the canonical-height argument, the weak-boundary construction, the attained fibre maximum, and the finite acceptance and composition rules. Its references prefixed "FP" point to the supplement's own numbering. The supplement contains the dependency map (Appendix A), canonical reduction details (B), transport data (C), source accounting (D), the complete cross-saturated cover and room interfaces (E), and a complete numerical certificate example (F). All three documents form the material to be reviewed together. This index connects their statements with the frozen mathematical inputs and executable proof objects.

Release `v1.0.1` indexes the unchanged proof data frozen in `v1.0.0`. This manuscript revision does not alter those certificates, their thresholds, or their identities, and it does not report a new whole-chain mathematical replay. A completed first-phase Lean project cold rebuild is documented separately in S8. The paper distinguishes three responsibilities:

1. universal analytic implications, including complete-matrix reconstruction and feasibility of the motions;
2. exact local verification of the supplied finite instances;
3. complete coverage and agreement of all inherited source domains.

The third responsibility includes recursive dependencies and cannot be inferred from file counts or hashes alone. The public verification guide specifies which component commands execute mathematics and which later commands compose previously checked results. The original v1.0.1 manuscript and index PDFs are archived at https://doi.org/10.5281/zenodo.22727368.

S1. The constant and attainment

`supplement/alpha_exact.json` contains all 62 ascending integer coefficients of the degree-61 polynomial and rational isolating endpoints $L < U$, with $U - L = 10^{-200}$. The published lower bound and its attaining equality family are due to Chen, Edelman, and Urschel. The identity record `evidence/v54/rho5_v54_alpha_identity.json` compares all coefficients and records the endpoint signs and the one-sign-change transformation $(1+t)^{61}P_{61}((4+5t)/(1+t))$. This identifies the same unique root in $(4, 5)$; it is not a separate global upper-bound proof.

The retained foundational reconstruction is in `supplement/foundations/`. R2 supplies coefficient and resultant reconstruction; R5 supplies rational Krawczyk isolation; R7 proves that the reconstructed matrix is actually completely pivoted and attains α. The readable attainment record is also copied to `supplement/R7_candidate_primal_cp.md`. Within the foundational directory the corresponding files are: `outputs/P5_coefficients_ascending.csv`, `outputs/P5_exact_verification.json`, `outputs/R2_free_CAS_reproduction_report.md`, `outputs/R5_candidate_interval_newton.md`, and `outputs/R7_candidate_primal_cp.md`. The actual entry points, preserving their original relative inputs, are `work/verify_p5_exact.py`, `work/candidate_interval_newton.py`, and `work/candidate_primal_cp_certificate.py`. They use `work/rational_interval.py` and the three polynomial files `work/sympy_recompute/p1.txt`, `work/sympy_recompute/p2.txt`, and `work/sympy_recompute/p3.txt`. The primal entry already rechecks the Krawczyk enclosure; running every entry separately is not required for every review.

S2. Early bounds, complete lift, and saturation

The scalar and saturation sources are `supplement/R8_scalar_root_bounds.md` and `supplement/R8_global_chart_coverage.md`. The complete two-step lift is R3, `supplement/foundations/outputs/R3_exact_lift_chebyshev_reduction.md`. The early bounds apply after normalization of the largest initial entry to one. The D/X saturation used in the article preserves a nonzero fifth-pivot magnitude and the complete earlier pivoting bands; zero fifth pivots are handled by the early bounds.

The preserved historical R8 saturation text is accompanied by `supplement/R8_SATURATION_ERRATA_V55.md`. A zero fifth pivot need not mean a zero fourth-stage tail, and the two full-matrix constructions are asserted equal only at a positive-product junction. The immutable historical text has SHA-256

`0ffdc19be82c27f3e7d660bb915ce1bcc051908c89c246fbea60cfe2b4ebecf0`.

The corrections, analytic proof map, and bounded checks are retained in `evidence/v55/ANALYTIC_AUDIT.md`, `evidence/v55/PROOF_DEPENDENCY_MAP.md`, and `evidence/v55/LOCAL_ACCEPTANCE_RECEIPT.json`. Those finite checks are not a replacement for the universal written arguments or a replay of the X and negative-D certificates.

S3. Canonical representative, reconstruction, and height

The canonical-height dependency has three distinct components. Their historical acceptance records must not be merged into a claim that the final composer re-executed all three.

Canonical representative (V24/T24). The original package is `RHO5_V24_Exact_Verification_2026-09-06.zip`. Its six complete exclusion trees contain 228,542 nodes and 114,274 contradiction leaves, with no open leaf. The package retains the mathematical theorem, model, exact acceptor, and nested earlier inputs. Its SHA-256 is

`2478b3d6c8d9189fd508e2af4695bb2086c6628d5049c75cef759bd3e455dfa8`.

Here T24 denotes the canonical target proved in this package. Its original statement is `RHO5_V24/THEOREM.md`, Sections 7–8, and the exceptional representative domain is defined in Section 2. The main article's Section 6.1 and FP Appendix B give the analytic chain explicitly: high-value budgets and signs; first- and middle-column capacities using common parameters; simultaneous tail reconstruction; an attained same-head wall maximum; compactness and minimum attainment at a prescribed target; and the qualified endpoint equivalence. The finite C19, difference, and exceptional-contact exclusions remain separately identified computational inputs. Their continuous source-to-model inclusions precede their use.

The safe-tail predecessor, retained among the nested V13 dependencies, has SHA-256

099eae4a478096c3d31a8ed10023ad2219f2879a0cb05fdd29ea64743020d247.

It contains TAIL_THEOREM.md, proof_certificate.py, certificate.json, and verify.py: thirteen fixed polynomial-positivity certificates and twenty-two symbolic identity checks. These are distinct from the later adaptive exclusion trees. The article states the needed canonical-domain reduction and finite positivity conditions; it does not assume the safe-tail implication for an arbitrary signed source.

To replay the six V24 exclusions together with the nested earlier acceptances, run the following in RHO5_V24/:

```
python verify.py --jobs 4 --full-chain
```

Omitting --full-chain runs only V23's algebra after the new V24 checks, and does not re-execute the V22/V20 exclusions. The V25 handoff restates mathematical inputs; it does not replace their proofs or the indicated finite acceptance.

Row-scale reconstruction (V25). The original nested object is RHO5_V26_Exact_Verification_REBUILT/V25_DEPENDENCY.zip inside RHO5_V26_Canonical_Global_Alpha_Exact_Verification_2026-09-07.zip. The SHA-256 of the nested ZIP bytes is

2ee17e8eb8ff5d21716675ce86eea2cc791dc2a9454d820c90bd481e164a47ca.

The universal equivalence is analytic. Its 28 symbolic checks and rational regressions test implementation and reconstruction; they are not a sweep of all parameter values.

Canonical exact height (V28). The preserved package is rho5_v28_exact.zip, with SHA-256

bec88339a158920d9dacb15341b02f359af83c4bf0e977721d9bdb2764fa0668.

All thirteen localization fronts were continued. The proof has 152,467 records and 208,274 contraction certificates; its three terminal boxes in the peak neighborhood are paid by the analytic and algebraic height module. The independent replay used the earlier V24/V25 mathematical inputs without rerunning every older tree. V27 entrance dictionaries and other earlier dependencies are retained where the frozen component manifests require them.

Additional upstream replay, 12 September 2026. The unmodified V24 entry with --full-chain was executed in a fresh directory. All 22 V20/V22/V24 exclusion trees passed: 1,275,818 nodes, 637,920 contradiction leaves, and no unresolved leaf, together with the invoked nested algebra and reconstruction checks. A separate fresh V28 run passed all eleven required modules, including all thirteen localization grafts and the remaining analytic and algebraic interfaces. The respective elapsed times were 862.85 and 44.94 seconds; the frozen static inputs were unchanged. Logs, identities, environment details, and precise scopes are retained in data/replay_receipts/. These are fresh executions of the frozen acceptors, not independently rewritten acceptors or a new full-theorem replay; the universal analytical source-to-model arguments remain separate mathematical responsibilities.

Readable height argument. Sections 6.1–6.3 of the main article and FP Appendix B expand the mathematical chain: qualified representative existence, necessary and sufficient row-scale reconstruction, complete localization, a constrained maximum, forced contact equalities, and the exact algebraic height. The equality pattern is a conclusion of the maximum argument, not an assumption on all physical sources. A common polynomial root or an isolated stationary point does not replace these steps.

The statements inherited by the representative theorem retain their full head budgets, positivity qualifications, and reconstruction ranges. The frozen older inputs supply those qualifications; the low-r certificate consumes the resulting bound through a separately qualified local transport. The dependency direction is canonical height to complete X, and then complete X to negative-diagonal safety. Final negative-diagonal coverage is not used to establish its own upstream canonical-height input.

Canonical-height replay interfaces. The V28 entry verify_v28_all.py checks the hash manifest, integer interval routines, rejection controls, and all required modules. The component verify_v28_coverage.py attaches thirteen certs/front_*.jsonl.gz payloads to the preserved partial localization tree. It verifies original and child boxes, contractions, conditional rows, and the three actual containments in the peak box. The resulting 152,467-record cover must not be confused with its incomplete earlier checkpoint.

The entry verify_v28_symbolic_bridge.py checks the 37-expression guard dictionary, while verify_v28_bridges.py checks 42 rational transport and qualification statements and their match with the Jacobian box. Under RHO5_V26_REBUILT/, verify_upper_tree_recovered.py accepts the 169-node upper-target tree, and verify_jacobian_recovered.py accepts the uniform nine-variable certificate. The recovered certificates are

equivalent repaired proof objects, not a claim to reproduce lost earlier bytes. The entries `verify_v28_carrier.py` and the rebuilt `verify_resultant.py` check the carrier identities and resultant against the complete defining polynomial. Root uniqueness in the article also follows from the separately established root in $(4, 5)$.

S4. The complete cross-saturated domain

The complete X certificate collection comprises the following archive objects. The table gives their exact filenames and replay entry points; commands are run from the named extracted package root. Individual archive instructions specify their input dependencies.

Domain and source	Archive filename	Exact entry
$0 < k \leq 2$, V31	`rho5_v31_exact.zip`	`python verify_all.py`
$k \geq 2.2$, Round43	`ROUND43_MINIMAL_REPLAY.zip`	`bash repro/replay.sh`, in `RHO5_CQG_ROUND43`
$2.15 \leq k \leq 2.2$, and strict high r for $2.1 \leq k \leq 2.2$, Round44	`ROUND44_MINIMAL_REPLAY.zip`	`python3 repro/replay_new.py .`
$2.14 \leq k \leq 2.15$, $r \leq k$, V33	`rho5_v33_exact.zip`	`python verify_all.py`
$2.1 \leq k \leq 2.14$, $r \leq k$, Round45	`RHO5_ROUND45_EXACT.zip`	`rho5_round45_exact/replay.py`
$2 < k \leq 2.1$, $r > k$, corrected V34/Round46	`RHO5_V34_HIGH_EXACT_FULL.zip`	`rho5_v34_high/replay_high.py`
$\gamma/2 \leq k \leq 2.1$, $r \leq k$, $F \geq \gamma$, Round47	`RHO5_ROUND47_ALPHA_FULL.zip`	`rho5_round47_alpha/replay.py`

The last three ZIPs have respectively 222,151,979, 131,835,895, and 367,175,732 bytes. The last package, comprising complete roots I and II, has SHA-256

`ddab67679b4faf3338beddaaea25dce99edfabeab937a0da07aa534c7a866de8`.

Round46 uses the corrected exact factor-graph implementation, not the superseded one. V36 assembles the unrestricted complete X theorem: `supplement/V36_THEOREMS.md`, with its full source at `rho5_v36_exact/THEOREMS.md`. The package instructions specify Python/SymPy, C++17, and Boost requirements. Their previously recorded acceptances are not represented as new runs performed in preparing this index.

Two distinct analytic entry rules. The main article's Section 6.4 and FP Appendix E state the complete strict- and weak-room predicates, the same-height strict-room reduction, the weak-room approximation at zero receivers and cofactors, and the qualified wall entrance. It distinguishes the room signs $A, B, d \leq 0$ from the wall certificate's stronger three negative signs. The local entrances do not cover every complete source; the original-root trees retain the other exits and boundary cases. The low-r package's `source/THEORY.md`, Theorem V35-local, and `source/FLOW_TUBE_THEOREM.md` describe the 22-coordinate X-to-room wall flow. It decreases $r + w$, then applies an actual signed tail permutation at $w = -r$; all sixteen strict room conditions are certified at the endpoint. Its upstream room bound uses the original two-face normalization (V10, Sections 2–6), the tail bound (V11, T11), the middle-face reduction (V12), and the independent canonical-height result in S3. The separate A4 exit invokes V27, Theorem 4.1, with all six weak cofactors. The 24-coordinate two-gap transport in V36, Section 6, instead preserves $r + w$ while eliminating the two cross gaps and consumes the complete X bound. Neither it nor the later V43 guard flow is an upstream assumption of the low-r certificate.

S5. Six-variable maximum in the negative diagonal domain

V36 gives the attained actual-tail maximum and qualified double-gap flow. V37 eliminates (β, p, e, r, s, t) at a fixed seventeen-coordinate framework, on each nonempty high-value fibre in the normalized $p \geq 1$ range. These

are same-source feasibility statements, without packet-rank assumptions. The preserved texts are `supplement/V36_THEOREMS.md` and `supplement/V37_SIX_VARIABLE_ELIMINATION.md`; the full V37 source is `rho5_v37_exact/theory/SIX_VARIABLE_ELIMINATION.md`. The complete framework model is `supplement/B17_FULL.json`, SHA-256

`39a65f2cbd6089dab3a403c6a72f365a02003737a641be86c3cf73cb70e76de3`.

The model contains canonical images of every hypothetical source above α, including the proper negative domain with $k \leq 2$. It is not restricted globally to a selected contact equation.

S6. Acceptance rules and their thresholds

Let $\gamma = 4132517/10^6 < \alpha$. Contradiction rules exclude complete sources with $F \geq \gamma$; height and qualified transport rules may instead establish only $F \leq \alpha$. An alpha-safe source need not be infeasible at the lower threshold. Exact nonnegative-weight, interval, contraction, and ancestor-image checks are part of acceptance, not merely discovery output or hash comparison.

The final local record has 47 old contradiction leaves, three alpha-safe leaves (H, A, B14), one V53 contradiction, one B15 canonical-source threshold exclusion, and two B16 canonical-source threshold exclusions. Each B15/B16 source covers both qualified conditional charts. The joint conclusion is alpha safety, not whole-anchor gamma infeasibility.

Within the complete B preservation archive, the actual mathematical interfaces include `deep_math.py`, `alpha_cover.py`, and the frozen depth500 verifier; source assembly uses `alpha_parent_cover.py` and `final_overlay_accept.py`. The final local mathematical producer `verify_joint_anchor.py` has SHA-256

`548de0119a5d69ff543ab4da4c5c7c5fd6a63e7f5707c4c5c70a7af4ad548bba`.

The source composer `final_overlay_accept.py` has SHA-256

`974b79b97732b02008207fec847a8a68796fcb939c5acf862340cb6c5acc0a13`.

The joint producer executes its frozen child verifiers; the later composer consumes identified earlier mathematical receipts and checks their source bindings. It does not re-execute all inherited mathematics.

Quantitative two-gap transport. The main article's Section 8.3 and FP Appendix C use the V36 `two_gap_certificate.json`, `verify_two_gap_certificate.py`, `gap_model.py`, and `support/x_local_model.py` from `rho5_v36_exact.zip`. They use twenty-three differentiation variables at fixed p, with bounds proved on the full twenty-four-dimensional box, including variation of p. All four boxes have the common parameter range

$$\frac{726212331}{500000000} \leq p \leq \frac{727012331}{500000000}.$$

The strict entrance condition applies to the whole accepted source region. An equality case of that entrance test is not accepted by this lemma unless an additional proved rule handles it.

Actual necessary-row example. The main article's Section 8.2 displays a genuine 49-row contradiction on a 67-coordinate lift. FP Appendix F gives the complete numerical calculation, while FP Section D.3 locates its full source data and ancestors. The complete rows and bounds are in `rho5_v53_exact/proof/TERMINAL_DETAILS.json`, inside the public `rho5-final-anchor-example.zip`. Its source binding is recorded in `SOURCE_CERTIFICATE.json` and `DEPTH19_COMPLETE_SOURCE_PROJECTION.json` in the same directory. Byte-identical copies of these three small inputs accompany this revision under `data/certificate_example/`; the corresponding V36 transport input is `data/two_gap_certificate.json`. These extracts make the displayed arithmetic and coordinate tables easy to inspect. Full source verification still uses the frozen archive and its acceptors.

S7. Complete negative-diagonal coverage and binding

The preserved base tree is `rho5_independent_acceptance_20260911_11/checkpoint/B17_ROOT.json` inside the historical-root archive. It has 2,102,766,386 bytes and SHA-256

`8e138dce9a65e49252a1f4714229b1ca95227943913f1eeefdf51714810a563c`.

Its exact cold replay accepted 749,693 nodes: 374,846 splits, 374,839 contradiction terminals, four alpha-safe terminals, and four explicit old-rule open terminals. The 750 enclosed B44 continuations have no internal open terminal. The cold record `rho5_independent_acceptance_20260911_11/COLD_REPLAY.json` has SHA-256

`55f3a9dd5090cfd9e21c479444d5cca1fc15fde2b5c22b8180bbd1c834156b37`.

The recorded Fraction replay ran on Linux with eight worker processes and took 22,815.35 seconds of elapsed wall time. That is a historical mathematical acceptance, not the runtime of the final composition.

The article's Q_1, Q_2, Q_3, Q_4 are the following actual old-root boxes; the historical indices are retained here to locate their payloads:

Article	Index	Original-root path	Paid responsibilities
Q_1	338726	`10011000010101010`	111/111
Q_2	435003	`10111000001100011000010111011`	240/240
Q_3	563285	`1101000101111`	159/159
Q_4	675104	`1110101001111`	56/56

The sum $111 + 240 + 159 + 56 = 566$ counts constituent sources inside these four original-parent obligations. It is distinct from the $465 + 4 = 469$ count of original parents. FP Appendix D explains the source hierarchy. The revision's `SOURCE_ACCOUNTING_566_TO_469.json` gives a record-by-record index into the four frozen parent receipts, together with the actual root bindings and original file identities. It is an index of existing records, not an additional mathematical acceptance. Relative paths inside a transformed parent must be interpreted through their full inherited source maps, rather than concatenated to root paths.

The last local subtree, at relative path `10011101`, covers 54 terminal positions and pays only the final one of the 240 responsibilities of parent 435003. The original-parent accounting is 465 old accepted parents plus these four completions, hence 469/469 and effective open count zero. The old root bytes still contain four O records; they are paid by the separate cover, not relabelled. The article combines the historical types from S6 into 48 contradiction terminals, three height or transport terminals, and three canonical-source threshold exclusions. These are the same 54 terminal responsibilities, not additional proof objects.

Final local proof. `evidence/joint/ANCHOR_COMPLETE.json` and `evidence/joint/COMPLETED_ANCHOR_LEDGER.json` record the B15/B16/V53 joint acceptance. The former, from the recorded 184.37-second run, has SHA-256

`648aa5322cfd518d1eaa549bf806b3daab67123078c2aa64909ba97831dd11d1`.

Last parent. `evidence/final_overlay/PARENT_435003_COMPLETE.json` has SHA-256

`6451a87db7383915b58bb3174581a4c81c38eca9a8aae32641d00083ad6cacb9`.

Actual root cover. `evidence/final_overlay/ROOT_COMPLETE_OVERLAY.json` has SHA-256

`43dbb784c871ecad4b6ff10047f39a2dad7a69a0b6eb392b40c9a56306e6ea34`.

Its 357 direct dependencies are enumerated in `FINAL_OVERLAY_DIRECT_DEPENDENCIES.json`. Recursive runtime and upstream theorem dependencies are additional objects; this list alone is not the complete proof archive.

Fresh negative-diagonal replay, 30 September 2026. The dated replay supplement contains `replay_engineering/EXECUTABLE_DEPENDENCY_MANIFEST.json`, the sealed 570-job plan covering 566 distinct targets, and `replay_engineering/replay_parents.py`, its mathematical driver. The unchanged executed plan and driver have SHA-256 values

`1bb4be62a66fe3570ea845c33ceb3df4e47f48d44899ea7f718a972eda50717f`
`bd1ea197b0342e095f7d53bd2f2d325270e92f1e967c40349fee3733bd70e53e`.

The receipt archive `FINAL_REPLAY_RECEIPTS_20260930.tar.gz`, under `replay_engineering/x_return/final_20260930/`, has SHA-256

`24f85c5ec1128409bdd1c15015d7f2227755f3ac61ef1f322270166ff40aefab`.

It contains 628 indexed payload files plus the export manifest itself (629 regular files). The four whole-anchor complements retain their specified open paths as intermediate obligations. The accepted target cover and actual-root binding yield effective open count zero; 570 successful producers do not mean 570 independently closed domains. The archive preserves the actual executed code, root and parent receipts, and failed first root attempt. Historical inputs copied under `fresh/joint/` remain historical inputs, not newly generated proofs. This lightweight archive does not include all certificate payloads. Its scope is the negative-diagonal finite cover relative to retained upstream analytic and complete X results, not a fresh full-theorem replay.

S8. Artifact collections and reproducibility scope

The manifest groups the proof objects into the following logical collections. Each collection groups the retained files by its mathematical role; the public asset names are given below.

Companion analytical collection. The relative `supplement/` and `evidence/` locations used in S1–S7 contain the analytical sources, selected exact inputs, and acceptance records. The foundational `work/` and `outputs/` inputs retain their relative structure under `supplement/foundations/`; their identities are listed in `FOUNDATIONAL_MINIMUM_MANIFEST.json`.

Canonical and complete X certificates. The archive objects named in S3–S4 include their frozen mathematical inputs and acceptors. Their manifests determine the nested dependencies.

Complete negative-diagonal certificates. The historical-root and final-cover archives below contain the base root, its exact runtime, and the four complete source compositions.

The two complete negative-diagonal archives are:

`historical_root.tar.gz` 2,189,588,699 bytes; SHA-256 `15ef795e85d0d295e92fdd629f313b9c284cc6519388011862aee3c1e18e3f73`.

`final_cover.tar.gz` 1,385,430,364 bytes; SHA-256 `1c9fb99d184eaf9d79b44d33826e4db8fbced8db41ee2b46ec2e1616bd12361c`.

Their retained copies have passed whole-archive hash checks, recorded in the preservation collection's `LOCAL_ARCHIVE_RECEIPT.json`. This verifies byte identity, not a further mathematical replay. The second archive retains the relative locations `rho5_deep500_all_open_20260910_08/`, `rho5_b16_v53_closure_20260911_18/`, `rho5_b16_20260911_17/`, and `rho5_cqg_continuous_20260910/run_01/runtime/models/B17_FULL.json`. The first contains the base tree and its `runtime/R54_SOURCE_MANIFEST.json`, listing 959 runtime sources, under `rho5_independent_acceptance_20260911_11/`. These are archive-relative locations. Frozen absolute-location bindings require an explicit replay mapping that retains every mathematical input identity and source check.

Final local source. The 18,292,611-byte final-anchor review bundle, SHA-256 `88a05ffe78a54d0c552aa6c5f2294e8614b7f78296efbbfb0cd5332322d21e15`, contains a standard-library-only short replay for Python 3.10 or later:

```
python -S restore_inputs.py
python -S verify_joint_anchor.py --v53-dir rho5_v53_exact \
  --b16-dir B_STRUCTURE_16_DELIVERY --output-dir replayed_anchor
```

The output directory must be empty or absent. The result pays the 54-leaf anchor only; its standalone whole-parent and whole-domain flags correctly remain false.

Complete negative-diagonal domain. From the extracted historical-root directory, the exact root entry is:

```
python -B -S runtime/r54_replay.py --tree checkpoint/B17_ROOT.json \
  --out REPLAY_ROOT_NEW.json --workers 8 --backend fraction --allow-open
```

The new output must not already exist. Its four expected O records are covered by the separately accepted parent proofs. `deep_math.verify_branch` and `alpha_cover.verify_cut` are mathematical interfaces; `alpha_parent_cover.py` and `final_overlay_accept.py` compose accepted sources without rerunning all the mathematics of their input receipts. A complete independent replay therefore includes both the parent mathematics and the

final source binding. The frozen 239/240 baseline must be preserved when attaching the last local proof, with a hash-bound replay configuration and fresh output locations. Python assertion optimization must not be enabled.

Public download and environment guide. The published inventory totals 4,494,382,372 bytes, including intentional asset overlap. This is not a transitive dependency-closure claim. `docs/DOWNLOADS.md` separates download size, extracted size and working-space estimates. The helper `scripts/fetch.py` verifies asset hashes and reconstructs the multipart root archive; downloading is separate from mathematical acceptance.

C++ components require a compatible C++17 compiler and the specified Boost headers. Python dependencies, working directories and migration constraints are given in `docs/VERIFY.md` and `docs/VERIFY.zh-CN.md`. The global theorem also uses S1–S5, including the complete X certificates, attained lower bound and analytic boundary arguments. In the published supplementary analysis archive `RHO5_PAPER_REVIEW_PHASE1_20260912.zip`, the root-level files `THEOREM_CHAIN.md` and `REPRODUCIBILITY.md` specify dependencies, entry points and scopes. Recorded component acceptances and source compositions are preserved; a tested, new-machine single-command replay of the entire theorem is not claimed. Publication and hash checks do not constitute that replay.

Ordered independent reproduction. The minimal dependency order is S1–S2 (constant, attainment and source reductions), S3 (canonical representative, reconstruction and height), S4 (all seven complete-X components), S5–S6 (the negative-diagonal source and qualified rules), then S7 (the original root, all four parent mathematics, and final composition). The S3 V24 command must include `--full-chain` when the nested exclusion acceptances are to be regenerated. S3's V28 entry is `python3 verify_v28_all.py` in the extracted `rho5_v28_exact/` package. S4 lists the seven component entries. The actual domain theorem is assembled only after all of its required components and qualifications have been checked.

The four-parent mathematical interfaces live in `rho5_deep500_all_open_20260910_08/` inside `final_cover.tar.gz`. The function `deep_math.load_protocol` loads the frozen depth-500 protocol; `deep_math.verify_branch` checks that a continuation is attached to its actual source and invokes its exact verifier with cross-checks. The function `alpha_cover.verify_cut`, after configuration with the frozen rule support, replays the source-to-cut mathematics, including the ancestor image. These are Python interfaces, not standalone command-line commands in their historical packages. The dated supplement's `replay_engineering/CURRENT_REPLAY_STATUS.md` provides the later plan–root–parents–gates–composition command sequence; `PATH_MAPPING.json` gives its explicit source mappings. These commands orchestrate the negative-diagonal component, with the retained upstream inputs and portability limitations described above. Their producers precede `alpha_parent_cover.py`, which binds their accepted records. The final composer is `rho5_b16_v53_closure_20260911_18/final_overlay_accept.py`. It checks the complete parent target sets and the actual original-root bindings; it does not execute every earlier mathematical producer. For parent 435003 its input is the frozen 239/240 basis plus the last complete anchor, not an arbitrary later mutable state file.

A fresh complete reproduction records fresh mathematical results for all the required sources before the final composition. If historical acceptance records are reused instead, report that reuse and its source-binding checks explicitly. The four preserved root O records are acceptable only as the specified intermediate result; a final unpaid target is a failed complete reproduction. Hash mismatches, missing recursive inputs, altered models or disabled assertions are also failures, not permitted environment adaptations.

Version identities. The version of the article, the version of a source checkout and the version of a certificate asset need not coincide. The v1.0.2 article revision preserves the frozen certificate identities indexed by the v1.0.1 checkout, including assets originally published under v1.0.0. Their earlier URLs and SHA-256 values remain intentional. The DOI https://doi.org/10.5281/zenodo.22727368 identifies the original v1.0.1 PDFs and must not be cited as the archival identity of this v1.0.2 revision. The expanded v1.0.2 manuscript and index are archived at https://doi.org/10.5281/zenodo.23057256. The present journal submission rearranges that material into the main article and FP, with this index updated to their navigation. That archival DOI identifies the expanded public version, rather than these newly rearranged submission files. The accompanying revision change log distinguishes exposition and verification-documentation changes from changes to proof data; the frozen mathematical inputs and their identities are unchanged.

Second-phase source review snapshot. The separately supplied B366 snapshot preserves 1,452 accepted Lean modules and their source/progress evidence. Its packaging supplement adds four previously compiled reuse-support sources, separately identified as `StrictWall.lean`, `WallBudgets.lean`, `TailFeasibility.lean`, and `TailCoordinates.lean` in `Rho5.PhaseTwo.Canonical.Representatives.WallMaximum.MinimumE`. The original sources and manifest remain byte-for-byte unchanged; the four supports are not counted as newly adopted modules. The supplement records their hashes and frozen Git identities. A static import check finds no

missing PhaseTwo source import; 57 directly imported Rho5 modules outside PhaseTwo remain absent. Project configuration, external dependencies and their full transitive closure are not supplied, so this is not a standalone cold-build archive. No new Lean compilation was performed for the supplement.

The package records additional checked lemmas and limited concrete certificate instances; it does not supply a complete kernel check of the large trees or discharge the two global safety hypotheses. Its module count does not measure the proportion of the full theorem formally proved. The historical "awaiting pinned Lean compilation" comment in `CommonRepresentative.lean` is preserved; the supplement supplies the later successful compiler receipt and coordinator adoption record, bound to that exact source. The supplemented ZIP has SHA-256

`3c4dcb433ad5af1deee9c2f18531e862110c62c501cdbe5c54ed2a9d06d97fc7`.

First-phase Lean project cold rebuild. The frozen project was rebuilt on Linux on 12 September 2026, starting with no compiled Rho5 project objects. All 634 product modules, three additional example/audit modules, and 47 named axiom checks passed. The toolchain was Lean 4.30.0 with Mathlib revision `c5ea00351c28e24afc9f0f84379aa41082b1188f`. Pinned third-party caches were reused and missing dependency modules were rebuilt; this was a project cold rebuild, not a new installation and rebuild of the entire dependency environment.

The accepted route used an explicit dependency-order driver with per-module Lean invocations. The originally documented `lake build +Rho5.PhaseOne:olean` command failed in the cold state; a configuration-only repair was not counted as a completed build. The retained records distinguish that failure from the successful alternative route. The final `#check`/`#print` output still has the type

`XGlobalSafety` $\longrightarrow$ `RootEndpointSafety` $\longrightarrow$ `rho5Trace = alpha`.

The 47 axiom lists agree with the frozen records. The only axioms appearing in them are `propext`, `Classical.choice`, and `Quot.sound`. This check does not discharge the two explicit safety parameters. The large tree and certificate stage was not replayed by this build.

The frozen project source ZIP has SHA-256

`ddb1e4300ed03464024f2a2a227cee060b6b38de96a02043f9233f0412e28f56`.

The 107,995-byte cold-build receipt archive has SHA-256

`c0ec1be2dc8b2aff26f987d13be34a20f0f2e92a4d2389523e9fe88c9c980415`.

The v1.0.1 public source repository preserves that archive and the completion summary under `paper/data/lean_cold_build/`. It preserves the source/object manifest, finalization and axiom/type logs, and failed-build evidence; it is not a standalone replay package and does not contain all module logs, compiled objects, dependency caches, or the successful build drivers.